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MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Academic Year: 2022-2023

Program: B. Tech / MBATECH Stream;

Year: I Semester: II

B Tech IT, CE, CS, EXTC, MXTC, DS, CSE-CS, AI, AIML, AIDS/ MBA Tech IT, COMP, DS, AI

Subject: Data Structures and Algorithms

Time: __3___ hrs (2:00 pm to 5:00 pm)

Date: 28 / 06 / 2023 .

No. of Pages: 3

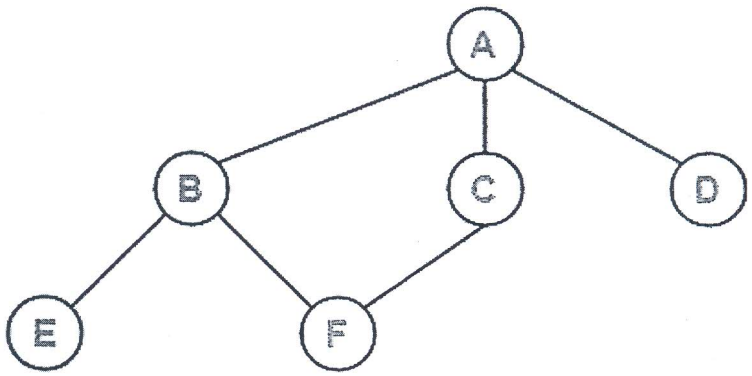
Marks: 100

Re-Examination (Batch 2021-2022)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. 1 is compulsory.
2) Out of remaining questions, attempt any 4 questions.
3) **In all 5_ questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	[20]
CO 1 SO 1 BL 2	a.	Explain Asymptotic Notations	[5]
CO 2 SO 1 BL 2	b.	Write a program to insert element at the beginning in singly Linked list	[5]
CO 3 SO 1 BL 4	c.	Explain different ways in which graph is represented in memory	[5]
CO 4 SO 1 BL 3	d.	Explain Binary search with help of an example and mention the pre – condition for Binary search	[5]
Q2			[20]
CO 2	a.	Write a function to insert and delete an element at the end in singly linked	[10]

SO 1 BL 3		list	
CO3 SO 1 BL 4	b.	Construct Binary Tree for given Inorder and Preorder sequence. Show all the steps while reconstructing the Binary tree. Inorder Traversal: {4, 2, 1, 7, 5, 8, 3, 6} Preorder Traversal: { 1, 2, 4, 3, 5, 7, 8, 6 }	[10]
Q3			[20]
CO 3 SO 1 BL 2	a.	What is the advantage of Threaded Binary Tree? Explain L-Thread and R-thread binary tree with the help of an example.	[10]
CO 1,4 SO 1 BL 2	b.	Which data structure is used in depth first search. Apply depth first search on the following graph and give sequence of traversal 	[10]
Q4			[20]
CO 2 SO 1 BL 2	a.	Explain Circular Queue data structure with the help of an example.	[10]
CO 4 SO 1 BL 2	b.	Write a program to implement linear search which will consider repetitive elements and give indexes of every occurrence of element	[10]
Q5			[20]

CO 2 SO 1 BL 3	a.	Apply Infix to postfix and Infix to prefix on the following expression using Stack A^B/D^E^F*G/H	[10]
CO 3 SO 1 BL 2	b.	Write a function to implement inorder, preorder and postorder traversal on binary search tree	[10]
Q6			[20]
CO 4 SO 1 BL 2	a.	Explain Breadth First Search with the help of an example. Apply insertion sort on following elements and show all the required steps 15,20,10,30, 50,18,5,45	[10]
CO 2,1 SO 1 BL 2	b.	Explain Binary Search Tree with the help of an example. Write a function to insert an element at beginning in doubly linked list.	[10]